

Hydrocarbons

A Level Chemistry

Arenes

Arenes 芳烃 are aromatic hydrocarbons, built on the **benzene** 苯 ring. The delocalised ring of electrons is stable and electron-rich, so benzene mostly reacts by **electrophilic substitution** 亲电取代—keeping the ring—rather than by addition.

Reactions of benzene and methylbenzene

Reaction	Reagents and conditions	Product
halogenation	Cl ₂ or Br ₂ , with AlCl ₃ or AlBr ₃ as a catalyst 催化剂	a halogenoarene 卤代芳烃 (aryl halide)
nitration 硝化	concentrated HNO ₃ + concentrated H ₂ SO ₄ , 25–60 °C	nitrobenzene
Friedel–Crafts alkylation 傅克烷基化	CH ₃ Cl + AlCl ₃ , heat	methylbenzene (adds an alkyl group)
Friedel–Crafts acylation 傅克酰基化	CH ₃ COCl + AlCl ₃ , heat	a phenyl ketone (adds an acyl group)
side-chain oxidation	hot alkaline KMnO ₄ , then dilute acid	benzoic acid 苯甲酸
hydrogenation 氢化	H ₂ , Pt/Ni catalyst, heat	cyclohexane

In the side-chain oxidation, the whole **side-chain** 侧链 (such as the –CH₃ on methylbenzene) is turned into a –COOH group, giving benzoic acid. In the hydrogenation, three molecules of H₂ add to the **benzene ring** 苯环 to make a saturated cyclohexane ring.

The mechanism: electrophilic substitution

Take nitration as the example. The acid mix makes the electrophile NO₂⁺. Then:

- the delocalised electrons of the ring form a bond to the electrophile, giving an unstable intermediate.
- an H⁺ is lost from that carbon, which restores the stable ring.

Substitution wins over addition because the **delocalisation** 离域 (aromatic stabilisation) of the ring is kept. Addition would destroy this stable system, so it is not favoured.

Side-chain or ring?

Where a halogen reacts depends on the conditions:

- with a halogen-carrier catalyst (such as AlCl₃) and **no** UV light → substitution in the **ring**.
- with UV light and **no** catalyst → free-radical substitution in the **side-chain**.

Directing effects

A group already on the ring decides where the next group goes—its **directing effect** 定位效应:

Group already present	Directs the new group to
–NH ₂ , –OH, –R	positions 2 and 4
–NO ₂ , –COOH, –COR	position 3